case 02 → or both ← case 01

Partially observable OR Nondeterministic (percepts determines which actions have actually occured)

Use percepts

they help in narrowing down the set of possible states it might be in.

Solution = action sequence

like [Suck-Right-Suck].

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if the environment is: Observable + Deterministic + Completely Known.

Trivial to solve the problem with previous algorithms.

1 Non determinism

using the function that returns the "set of possible outcome states" (after generalizing the notion of transition model)

We use AND nodes and consider all nodes (each alternative)

Leads to: AND-OR search.

+ Generalize the notion of a solution to the problem: because there is more than one solution for the problem

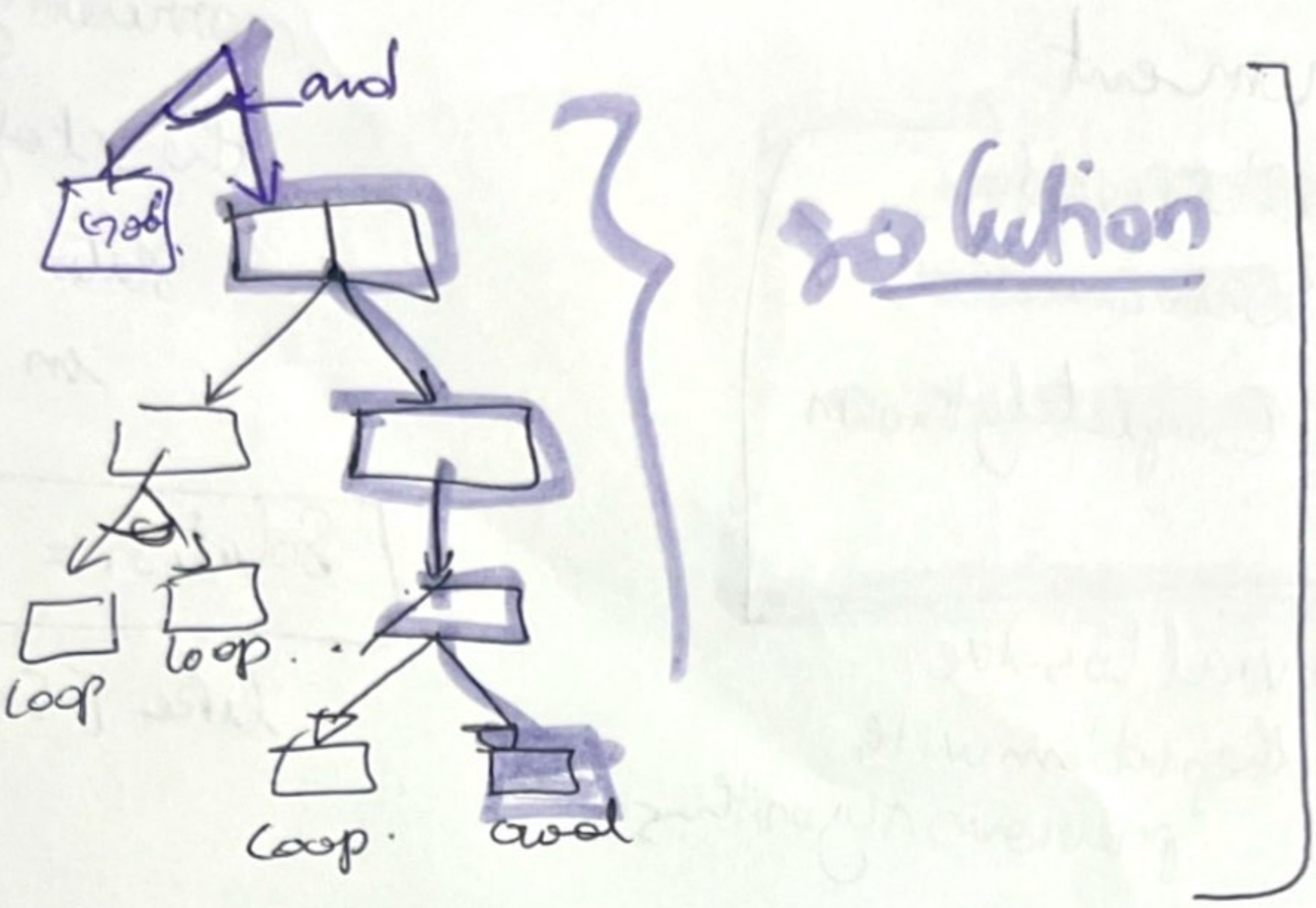
→ We need a Contingency Plan

like: [Suck, if State = 5 then [Right, Suck] else []]

So solutions for nondeterministic problems can contain nested if then else statements.

A solution for an AND-OR search problem is a Subtree that:

(1) has a goal or a Loop node at every leaf
(2) specifies one action at each of its OR nodes
(3) Includes every outcome branch at each AND node



Key aspect:
- Dealing with cycles. that rises in nondeterministic problems.

Current is identical to a state on the path from the root then it returns with failure
means that there is a cyclic solution. because if a noncyclic one is there it must have been reached before.

- check code slide 36

AND-OR graph can be explored by BFS or best first. and the heuristic must be modified to estimate the cost of a contingent Solution rather than a sequence!

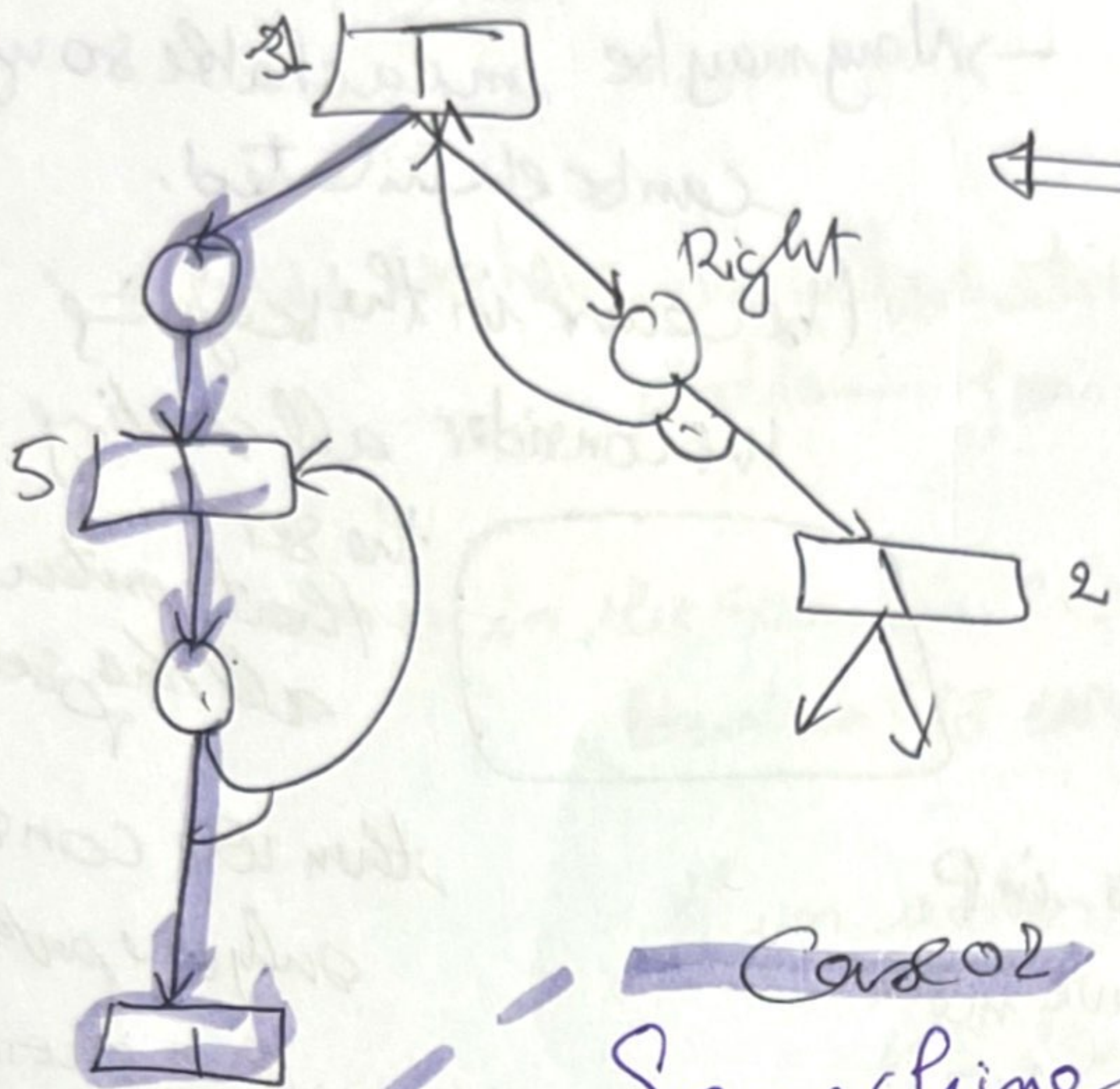
> there is an analog A* algorithm for finding the optimal solution.

Ensuring that the algorithm terminates in every finite space because every path must reach a goal / dead end or a repeated state

* When there are No acyclic solutions from a given state the AND-OR-GRAPH-SEARCH would return with failure.

* If there is a cyclic Solution (which is to keep trying until it works)
  → add a label to denote portions of the plan.

* This cyclic solution can be: [Suck, L1: Right, if State = 5 then L1 else Suck].



⇐ { ⇔ while (state ~~= 5~~) do right

Case 02

Searching with Partial Observations:

→ Searching with no observations (Sensorless problem) → can be useful sometimes
+ Conformant problem

→ Key Concept is: belief state

physical state space
⇓
Space of belief states
here the problem is fully observable! because the agent knows its own belief state

■ When a sequence of actions always takes to ~~the go~~ a given state S
We say that the agent can coerce the world into state "S"

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Since $\boxed{\text{no percepts}}$ at all $\Rightarrow$ $\boxed{\text{No contingencies to plan for}}$

even if the environment is nondeterministic!!!

Construction of the belief state search problem:

① Belief state: every possible set from the physical states

Power set $\rightarrow 2^N$ states in belief state if the physical has N states

$\rightarrow$ Many may be unreachable so up to 2^8 in the example can be eliminated.

(because in the begining we consider all starting states

the set that contains all the p states

then we consider only one path from one choice!

② Initial state:

Set of all states in P.

(Can have more inputs).

③ Actions

- if one is dangerous somewhere $\rightarrow$ take $\cap$
- otherwise $Act(b) = \bigcup_{s \in b} Actions_P(s)$

④ Transition model

$\rightarrow$ prediction step: generating the new belief state after the action is applied

④ $b' = PREDICT(b, a)$

deterministic actions

b' is never larger than b.

Non deterministic actions.

- $b' = Result(b, a)$

$= \{s' \mid s' = Result_P(s, a) \text{ and } s \in b\}$

$= \bigcup_{s \in b} Result_P(s, a)$

it can be larger than b.

- percepts is also a set.

⑤ Goal test — belief state does satisfy the goal if all physical states in it satisfy the goal.
(it may achieve the goal earlier and doesnt notice!)

⑥ Path cost.

for our vacum world sensorless cleaner we have 12 reachable belief states out of 2^8 possible.

→ We can have an automatic construction of the belief state problem formulation from the def of the physical problem.

⑤

→ in the search, We test if the stats are identical to existing one.

if an action sequence is a solution for the belief state b → then it is also a solution for any subset of b.

EXTRA Level of pruning that will dramatically improve the effeciency of solving the problem.

→ discard the path reaching {1,3,5,7} if {5,7} is reached already been generated

→ if {1,3,5,7} has already been guarenteed and found to be solvable, then any subset is also guaranteed to be so!

→ even with this: sensorless problem solving is seldom feasible in practice.

→ the difficulty is in the size of each belief state

Representation of B-States

English Words

Avoid by the standard algorithm and look inside using.
incremental belief-state search
that builds up the solution one physical state at a time.
- If a state does not work go back and find a different path.

Very effective.

Deterministic
- PERCEPT(S) function returns the percept recieved in a given state

Nondeterministic: returns the set of percepts.

Fully observable problems:
Special case in which
→ PERCEPT(S) = S for every state s
while sensorless problems are a special case in which percept(S) = NULL

Partial observation ⇒ several states would have produced any given percept

giving the initial percept ⇒ the initial B state will be different for the agent.

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Transition Model for partially observed problem: Ac1

from BS1
- step1
- step2
- step3

act…

BS2

① Prediction

② Observation prediction

③ Update

I go and update the ones that matches what you have predict.
→ update the prediction state

ALL together:

RESULTS(b,a) = { bo : bo = UPDATE(PREDICT(b,a), o) and o ∈ POSSIBLE_PERCEPTS(PREDICT(b,a)) }